

Cellular and Medical Immunology

Innate Immune Response

Integrated Biomedical and Clinical Sciences OMFS513

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Learning Objectives

1. Describe the pathogen recognition patterns.
2. Describe the general features the innate immune system.
 - a. Describe physical and chemical barriers
 - b. Describe differences between PAMPS, DAMPS and antigens
 - c. Describe phagocytic cells
 - d. Describe components on the complement system
3. Describe outcome of the innate immune response

Outline

- Pathogens
 - Pathogen recognition system.
- Innate immune response.
 - Phagocytic cells
 - NK cells
 - Complement system
- Outcome of innate immunity
- Kinetics of inflammatory response

Basic terminology

Allergen: An antigen that causes allergy.

Antigen: Any molecule that can be recognized by an antibody to produce an immune response.

Adjuvant: Substances that can enhance the immune response to antigen. Adjuvants are important for vaccine production.

CD (Cluster of differentiation): A designation used to catalogue surface molecules on leukocytes.

Cell-mediated immunity: This refers to T-cell mediated immune response.

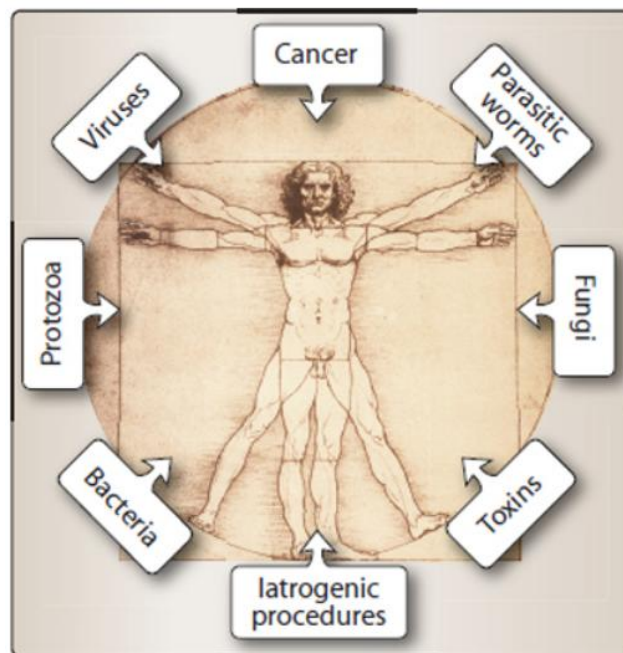
Humoral immunity: refers to antibody mediated immune response. It is derived from the term “humors” which mean body fluids.

Clone: identical cells derived from the same progenitor.

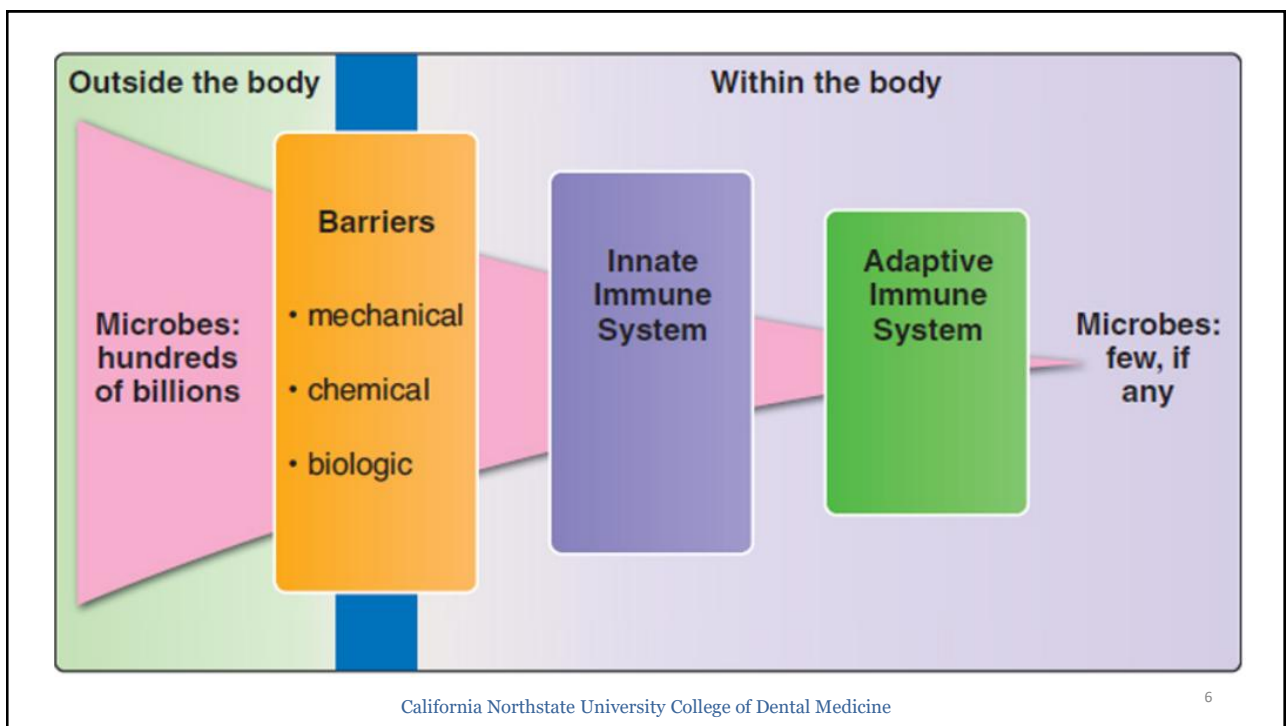
Immunogen: Any molecule that can produce an immune response.

Immune complex: A complex formed by an antibody bound to an antigen.

Lymphadenopathy: Enlarged lymph nodes.

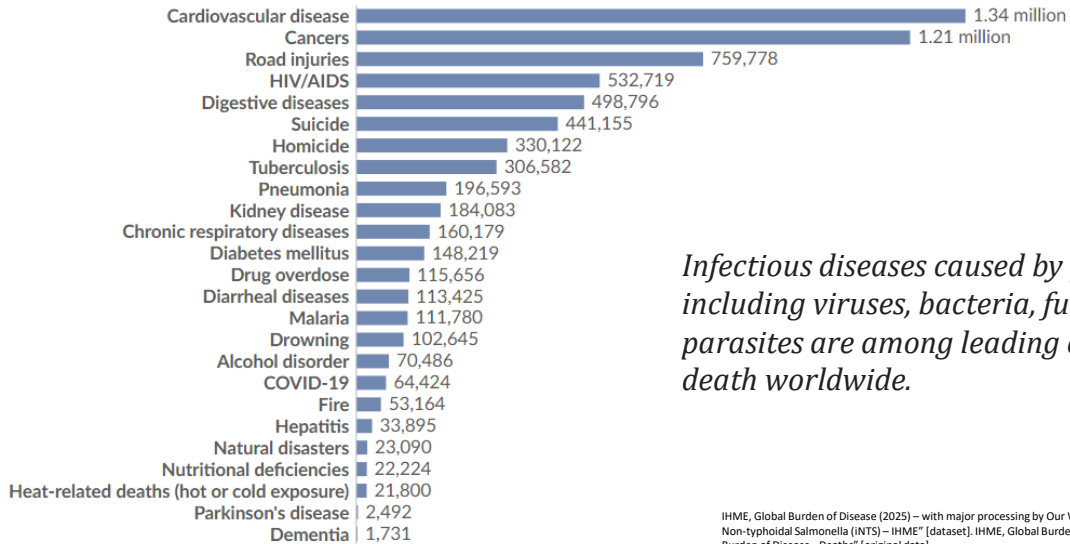


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Causes of death in 15- to 49-year-olds, World, 2023

The estimated annual number of deaths from each cause, among people aged 15–49 years old. Estimates come with wide uncertainties especially for countries with poor vital registration.



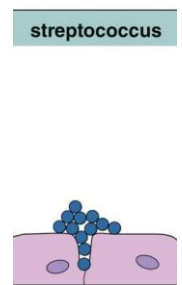
Infectious diseases caused by pathogens including viruses, bacteria, fungi, and parasites are among leading cause of death worldwide.

IHME, Global Burden of Disease (2025) – with major processing by Our World in Data. "Invasive Non-typhoidal Salmonella (INTS) – IHME" [dataset]. IHME, Global Burden of Disease, "Global Burden of Disease - Deaths" [original data].

Pathogen Replication

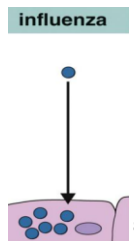
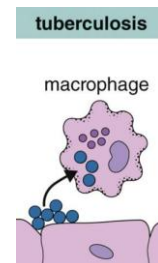
a) Extracellular pathogen

Replicate in the interstitial fluid without entering the cells. Examples include helminth, several fungal species, bacteria like *Staph. aureus*, *Strept. pyogens*, certain *E. coli*, *P. aeruginosa*, *V. cholera*.



b) Intracellular pathogen

Replicate in the host cell and manipulate its metabolic machinery to reproduce. Include viruses, few fungal sp. and certain bacterial sp. (*Mycobacterium tuberculosis* and *chlamydia*).



The immune response is modulated to efficiently fight each type of pathogen.

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What happens if bacteria can multiply freely

Bacteria doubling time is in minutes. A liter of bacterial culture can have as 3-4 billion bacteria.

How are Pathogens Recognized by the Immune System ?

PAMPS and Antigens

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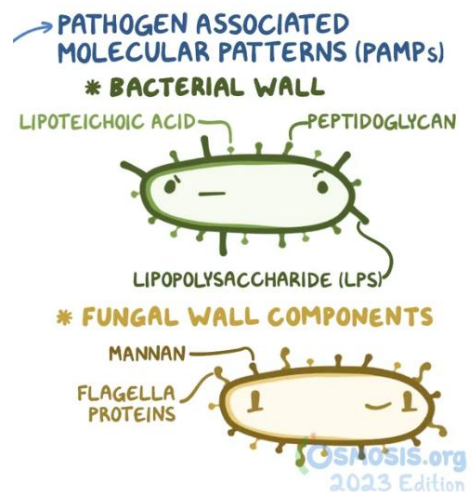
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Pathogen-Associated Molecular Patterns

PAMPs (aka MAMPs) are expressed by pathogens but not expressed by the mammalian cells.

PAMPS constitute a diverse set of microbial molecules that are expressed as a part of their life cycle. PAMPS can include pathogen-specific carbohydrates, lipoproteins or nucleic acids.

Detecting PAMPS signals immune cells to eliminate the intruding pathogens.



MAMPS= microbe-associated molecular patterns

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Pathogen	Pathogen-specific PAMP/MAMP
Bacterial products	Lipopolysaccharide (LPS), Mannose-containing carbohydrate, cell wall glycolipids, pilin and flagellin
Viral products	Viral genome (dsRNA, ssRNA).
Fungal products	Chitin (polysaccharides in the fungal cell wall), Zymosan (glucan on the surface of fungi).
Parasite products	Chitin (parasite cell wall), Schistosome egg antigens

Damage-Associated Molecular Patterns

DAMPs are endogenous chemical molecules that are released from damaged or dying host cells. DAMPs are also known as alarmins

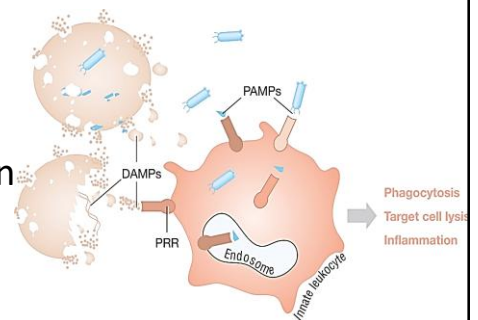
Source	DAMPs
Reactive oxygen species	Hydrogen peroxide.
Nucleic acid	Cell chromatin including fragments of genomic DNA.
Metabolic products	Uric acid, cholesterol and saturated fatty acids.
Stress molecules	Heat shock proteins, S100 family of calcium binding proteins.

Pattern Recognition Molecules

Pattern recognition molecules (PRMs) are sensors of innate immunity that recognize PAMPs/DAMPs.

PRMs can be **cell associated** (pattern recognition receptors, PRRs) or **extracellular**.

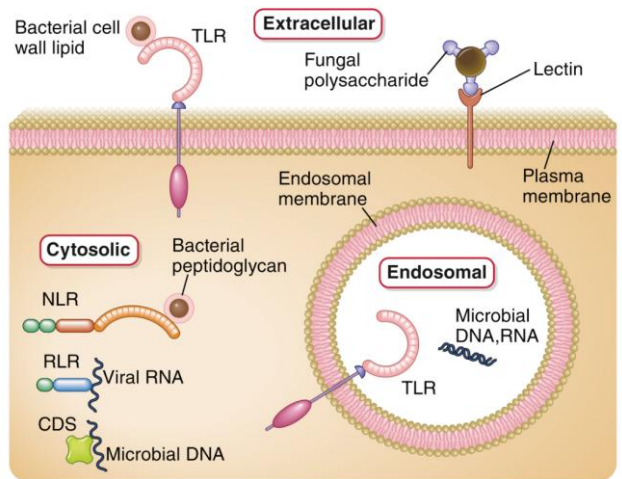
Following detection, innate immunity initiate a cascade of events leading to the destruction of the target pathogen.



Each PRR can identify specific PAMP/DAMP and generate an appropriate immune response

A) Cell associated or PRRs

1. Cell surface (membrane bound):
 - Toll-like receptors: (1,2,4-6 to detect bacterial cell wall components). (**LPS**, peptidoglycan All of TLRs are present on phagocytic cells, and some are also expressed on DCs, mast cells and B cells. Activation of TLR lead to phagocytosis and release of cytokines.
 - C-type lectin receptors (CLRs) for glucan.

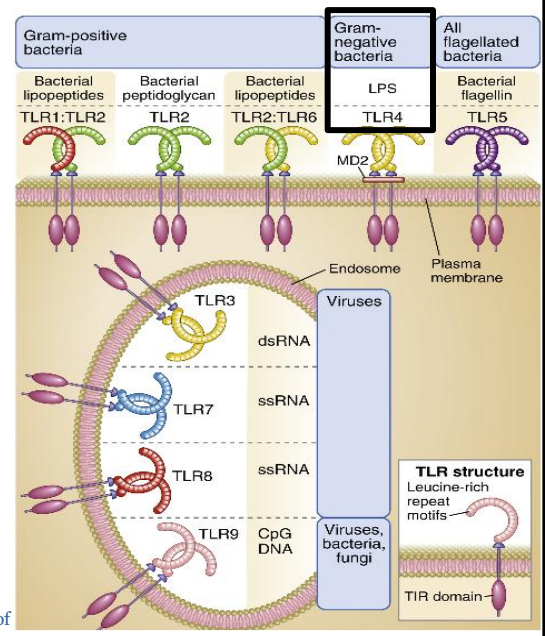


2. Cytoplasmic:

- Nucleotide oligomerization domain (NOD)-like receptors (NLRs) for bacterial cells wall, urate crystals, lysosomal damage, etc.
- Retinoic acid-inducible gene-I-like receptors (RLRs) for viral RNA

3. Endosomal:

- TLRs 3,7-9 to detect viral and intracellular PAMPs. TLR3 deficiency is can increase susceptibility to HSV infection. TLR7 variant is associated with higher viral load and disease progression in HIV.



B) Soluble PRMs

- Pentraxins family: example C-reactive protein (CRP; acute-phase protein produced by liver). Bind pathogens and activate complement system (classical pathway).
- Complement: C3b; facilitate opsonization and microbial cell lysis.
- Others.

NOD-like receptors and inflammasome

- Following activation, certain NLRs can interact with cytoplasmic proteins to form **“Inflammasomes”**.
- Inflammasome generate active inflammatory cytokines IL-1 and IL18 to promote the inflammatory response. This aids in an efficient innate immune response.
- Inflammasome activity was observed in inflammatory, neurodegenerative and autoimmune conditions.
- In development: pharmacological agents that target the inflammasome activity or NLR inhibitors.

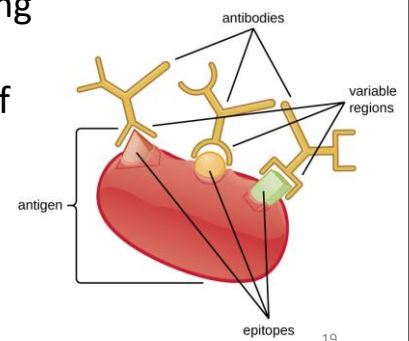
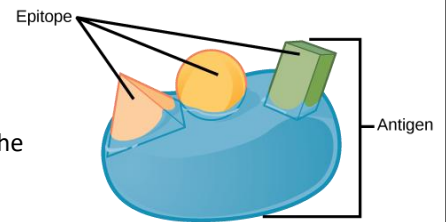
Antigen

An antigen are proteins/peptides that can elicit a specific immune response.

Each antigen has an epitope, (aka, antigenic determinant). An epitope is the immunological active region that can stimulate the immune response.

Types:

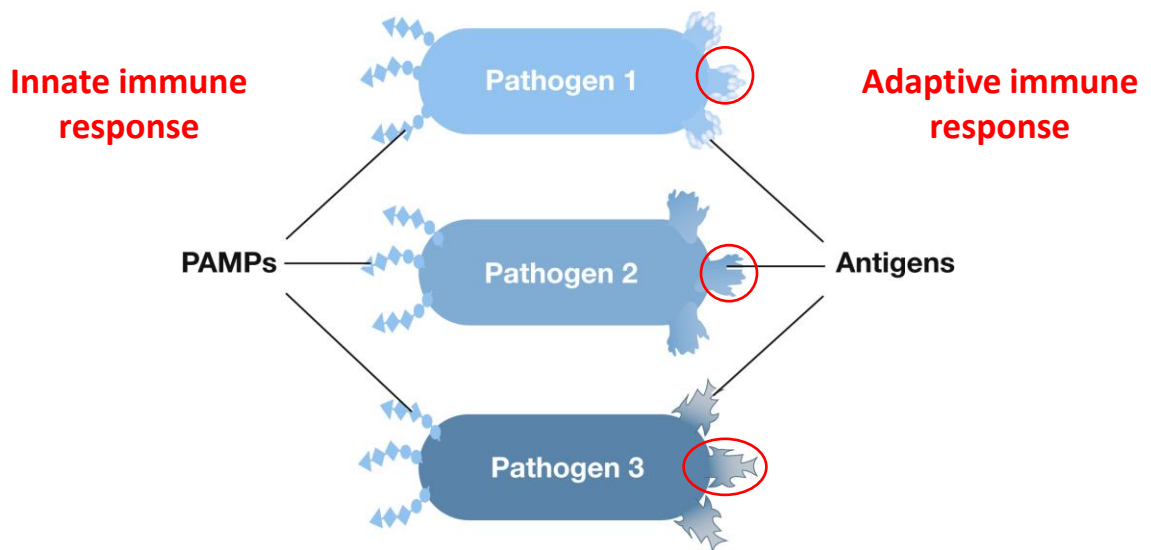
- 1- Exogenous antigen, present on pathogens invading the body
- 2- Endogenous antigens, found within the cytosol of human cells. For example, viral proteins, proteins from intracellular bacteria, and tumor antigens.
- 3- Autoantigens, host proteins that can induce a self-immune response.

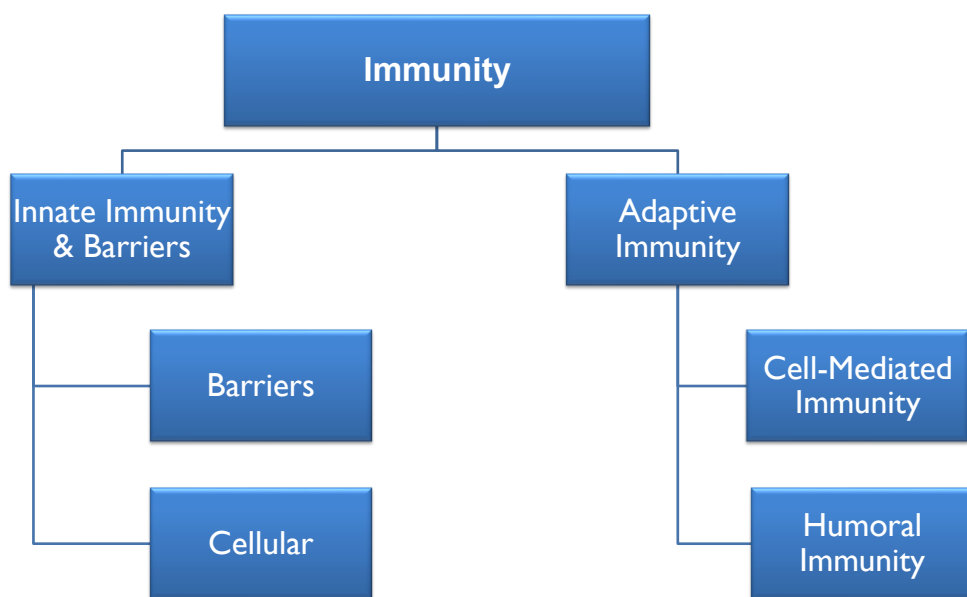


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PAMPs vs Antigens





Functions of the Innate Immune system

- 1- The innate immune system maintains physical and chemical defenses.
- 2- Innate immune responses prevent and eliminate infection. Deficiencies can lead to -----
- 3- Innate immune responses against resistant microbes aim to control the infection until adaptive immune responses are activated.

4- Innate immunity eliminates damaged cells and initiates the process of tissue repair. Activated by the recognition-----.

5- Innate immune responses stimulate and modulate ----- immune responses to increase efficiency of immune system.

6- Innate immunity can provide indicative signals to the adaptive immune system about the type of ----- . How?

Different components of the innate immune system react in distinct ways to different microbes (For example, extracellular bacteria versus intracellular viruses).

Innate Immunity

1- Non-inducible innate immune system:

Anatomical barriers against infection which can be both physical and chemical.

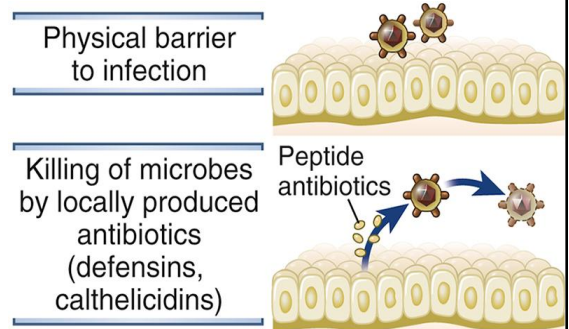
- a) Physical barriers.
- b) Chemical barriers.

2- Inducible innate immune system

Cellular responses to an infectious agent that overcame the physical and chemical barriers. It includes phagocytes, NK cells, complement system.

Physical and Chemical Barriers

- **Physical barriers:** Skin and mucosal membranes, BBB.
- **Chemical barriers:** secretions or chemicals released at the physical barriers with antimicrobial effect.
 - Defensins are produced in the s. intestine, colon, respiratory mucosa and skin by several cells. Defensins can be directly toxic to microbes and help in the activation of inflammatory response to microbes.



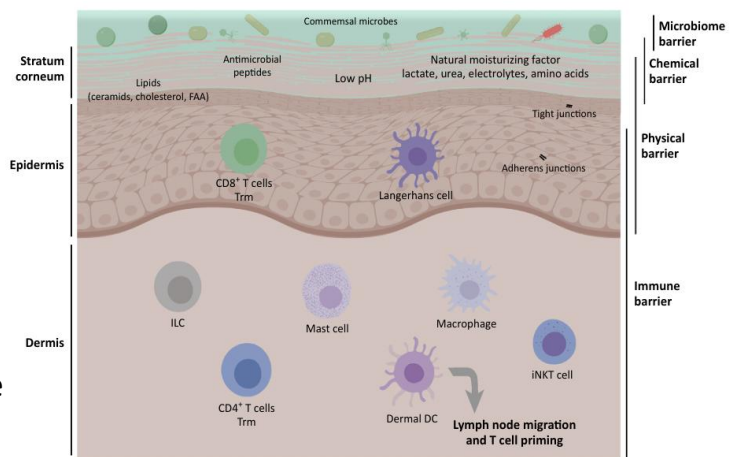
Cutaneous Barrier: Skin

Microbiome barrier: diverse microbial communities, prevent colonization by pathogenic species

Chemical: acidic surface pH and secreted compounds (defensins & lysozyme).

Physical: Stratum corneum (dead, keratinized cells) & tight junctions & others.

Immune: variety of resident immune cells populating the epidermis and dermis.



[Cutaneous Barriers and Skin Immunity: Differentiating A Connected Network: Trends in Immunology](#)

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Trends in Immunology

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Oral mucosa and saliva

Microbial: Oral microbiome. Change by age, health status and habits. Dysbiosis linked to oral and systemic complications.

Physical: Stratified Squamous epithelia, gingival epithelia keratinocytes with PRRs (TLRs).

Chemical: Antimicrobial peptides (including defensin), cytokines and salivary immune defense (pH, salivary IgA; mucins (to block microbial adherence), lysosomes, and others.

Immune: Several WBCs population are present in saliva (neutrophils are common) + various innate and adaptive immune components to prevent pathogen entry.

Barrier	Properties	Function
Skin	Keratinized epithelium	Dry surface inhibits bacterial reproduction; keratin layers and tight cell–cell junctions prevent microbial penetration; sloughing of cells and high turnover rate prevent colonization
	Lactic acid and fatty acids from sweat and sebaceous glands respectively	Low pH inhibits bacterial growth
Mucosal tissue	Secreted mucins	Form mucus with water; coagulates particulate matter, preventing attachment and facilitating ejection
	Ciliated epithelium (e.g. in trachea)	Works with mucus layer to physically eject particulate matter
	Flow of air (airways) and fluid (e.g. in the urogenital tract)	Prevents attachment and washes out microbes
Gastrointestinal tract	Acid environment in stomach (also in kidney and bile)	Low pH inhibits bacterial growth
	Digestive enzymes	Inhibit bacterial growth
Paneth cells in small intestine	Antimicrobial peptides, lysozyme and phospholipase	Destroy bacterial cells

Whawell, Simon, A. and Daniel W. Lambert. *Basic Sciences for Dental Students*. Available from: VitalSource Bookshelf, Wiley Global Research (STMS), 2017.

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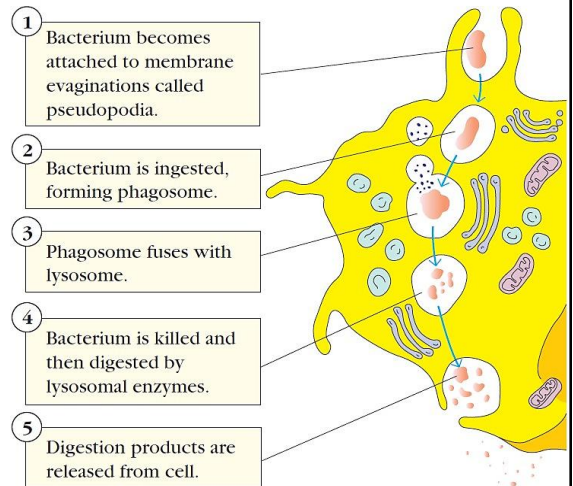
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Artillery of Inducible Innate immunity

- 1- Professional Phagocytes
- 2- Natural killer cells
- 3- Complement system
- 4- Cytokines

Phagocytosis

- Phagocytosis= cellular uptake of microbes and digestion via lysosomes.
- By macrophages, neutrophils, and dendritic cells in tissues, as well as monocytes in the blood.
- Active process (=----- use).
- Initiated with the polymerization of the actin cytoskeleton, and the development of membrane extensions around microbes, leading to their internalization, and the formation of phagosomes.



Activation of Phagocytosis

Direct binding of PAMPs to PRRs on phagocytic cells. This binding also activates the production of inflammatory cytokines and antimicrobial proteins.

Indirect or opsonization: phagocytes recognize soluble proteins, (named opsonins), that are bound to microbial surfaces.

Immunoglobulin G, or the complement protein (C3b), are examples of opsonins that coat the outer surface of microorganism

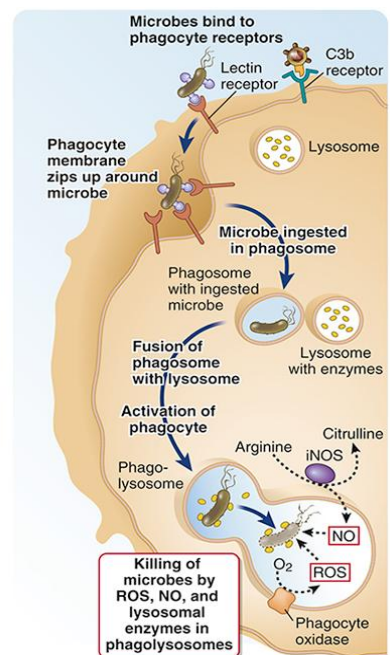
Opsonins are recognized by membrane opsonin receptors on phagocytes leading to the activation of phagocytosis.

Threat Elimination

1- Phagosomes fuse with lysosome yielding a phagolysosome. Contain antimicrobial agents (defensins), low pH, and hydrolytic enzymes (like proteases) which destroy the microorganism.

2- To eliminate the phagocytosed microbe, phagolysosome can mediate a respiratory burst (oxidative burst) by:

- a) Proteolytic enzymes neutrophils & macrophages.
- b) ROS: mainly neutrophils (less in macrophages).
- c) Nitric oxide by macrophages

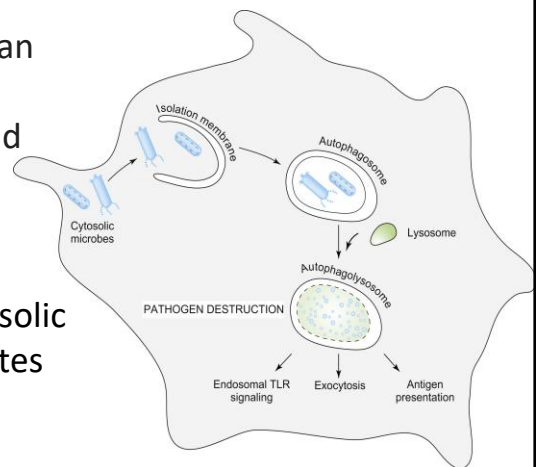


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Autophagy is another mechanism to clear intracellular pathogens

Autophagy is a self-degradative process that can be used to maintain homeostasis through removing misfolded or aggregated proteins and clearing damaged organelles.

In leukocytes, macro-autophagy involves the formation of autophagosome around the cytosolic microbe and fusion with lysosome. It contributes to clearing intracellular pathogens.



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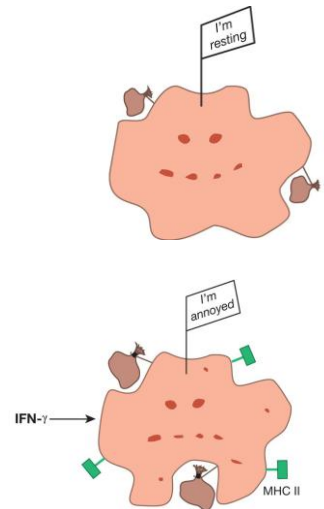
Macrophages: three stages of activation

1- Resting state:

Macrophages primarily function to collect dead cell debris.

2- Primed or activated state:

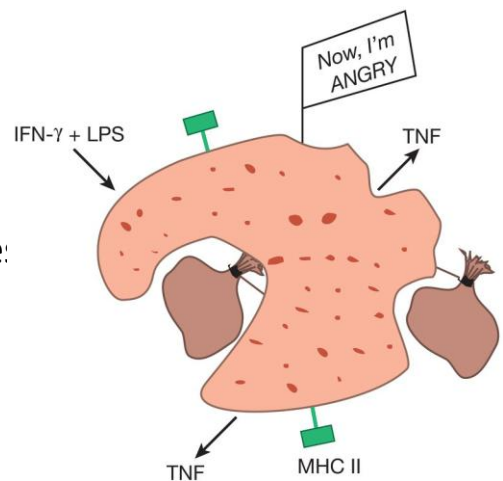
Following pathogen recognition or activating signals from IFN- γ , macrophages engulf invaders, upregulate the expression of class II MHC on cell surface and function as an antigen presenting cells.



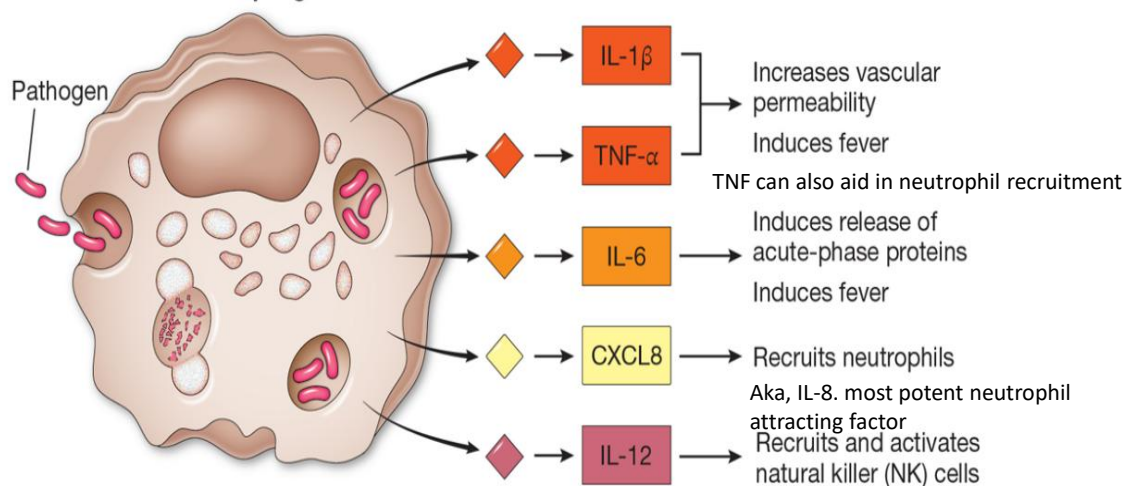
3- Hyperactivated state

- Has increased number lysosomes,
- Has increased production of ROS and nitrogen species.
- Can dump the contents of its lysosome: onto multicellular parasites.

The activation state of a macrophage depends on the type and the strength of the activation signals that it receives.



Macrophages: Inflammatory Cytokines



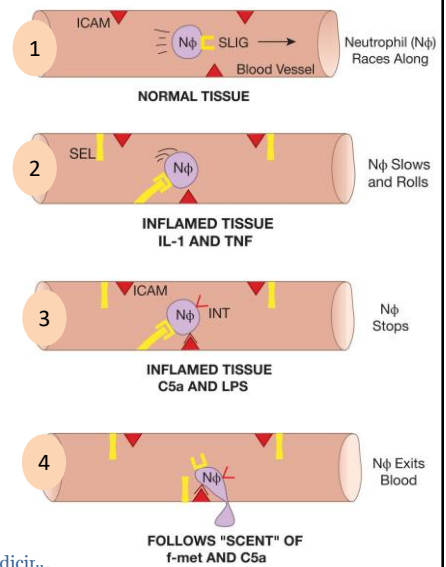
Macrophages secrete inflammatory cytokines that facilitate migration of innate immune cells toward the site of infection

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Neutrophils (aka foot soldiers)

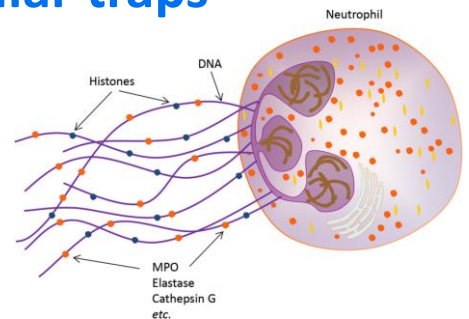
- Professional phagocytes. No antigen presentation.
- Represent about 70% of the circulating WBCs.
- Dead neutrophils are component of pus.
- Neutrophils express CXCR1 and CXCR2 receptor that recognize CXCL8 chemokine.
- Recognition of CXCL8 produces changes in neutrophil to enhance adherence to blood vessels wall, and travel to infection site.



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Neutrophil extracellular traps

- In addition to phagocytosis, neutrophils can kill microbes by the neutrophil extracellular traps (NETs).
- It involves the release of neutrophil DNA and antimicrobial granule contents to form a web structure that can trap and kill bacteria and fungi.
- This process leads to neutrophil death (aka, NETosis).
- Excessive NET formation can contribute to autoimmune and other inflammatory diseases.



Neutrophil- T cell interaction

- Can modulate T cell responses via
 - Indirect activation via DCs and cytokines/chemokines
 - Indirect suppression via ROS and release of arginase.

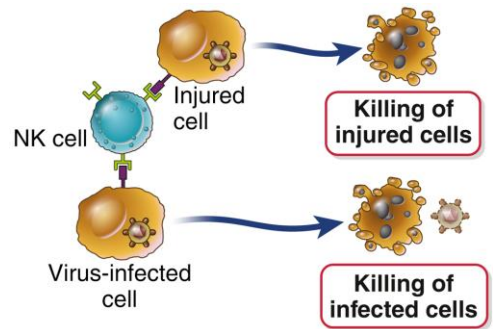
Macrophages vs neutrophils

- Both carry out phagocytosis.
- Neutrophils are the first leukocytes to be recruited to the inflammation site.
- Key differences:
 - Neutrophils are short-lived (1-2 days), has a wide range of anti-microbial polypeptides and readily produce ROS. More mobile.
 - Resident macrophages are long-lived with self-renewal capacity, less microbicidal and cytotoxic. Therefore, they act as tissue sentinels of the immune system.
 - Antigen presentation and NETosis

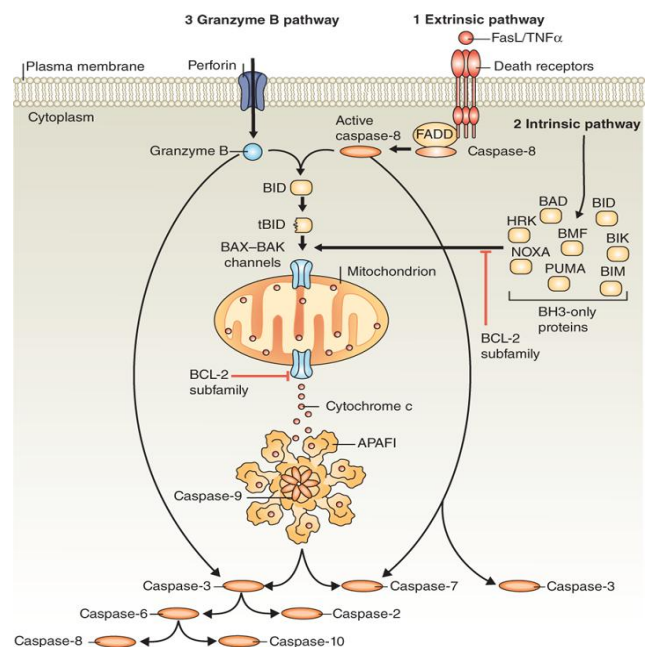
Artillery of inducible Innate immunity

- 1- Professional Phagocytes
- 2- Natural killer cells**
- 3- Complement system
- 4- Cytokines

- Natural killer cells are part of ----- immune response. CD16 positive.
- Cytotoxic cells that help destroy virus-infected cells, intracellular bacteria, and some cancer cells.
- Recognition of viral infection by PRRs triggers the expression of interferons (IFN- α and IFN- β).
- Interferons induce NK cell proliferation.
Function to prevent viral replication



- Activated NK cells release granule proteins:
 - Perforin: facilitates the entry of other granule proteins (granzymes) into the cytosol of target cells.
 - Granzymes are proteolytic enzymes that initiate signaling cascade that causes apoptosis of target cells.

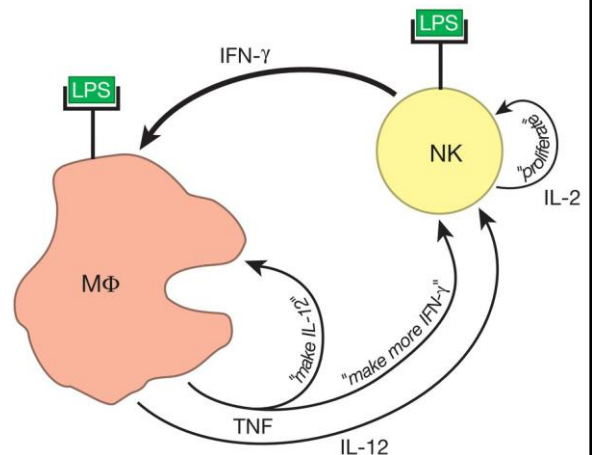


Source: Howard M. Reisner
Pathology: A Modern Case Study, Second Edition
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Macrophage-NK interaction:

- NK cells are activated by IFN- α of virus-infected cells or LPS of bacteria.
- NK cells secrete IFN γ that activates macrophages.
- In return, macrophages secrete IL-12 and TNF that activate NK cells and other immune cells.



Artillery of inducible Innate immunity

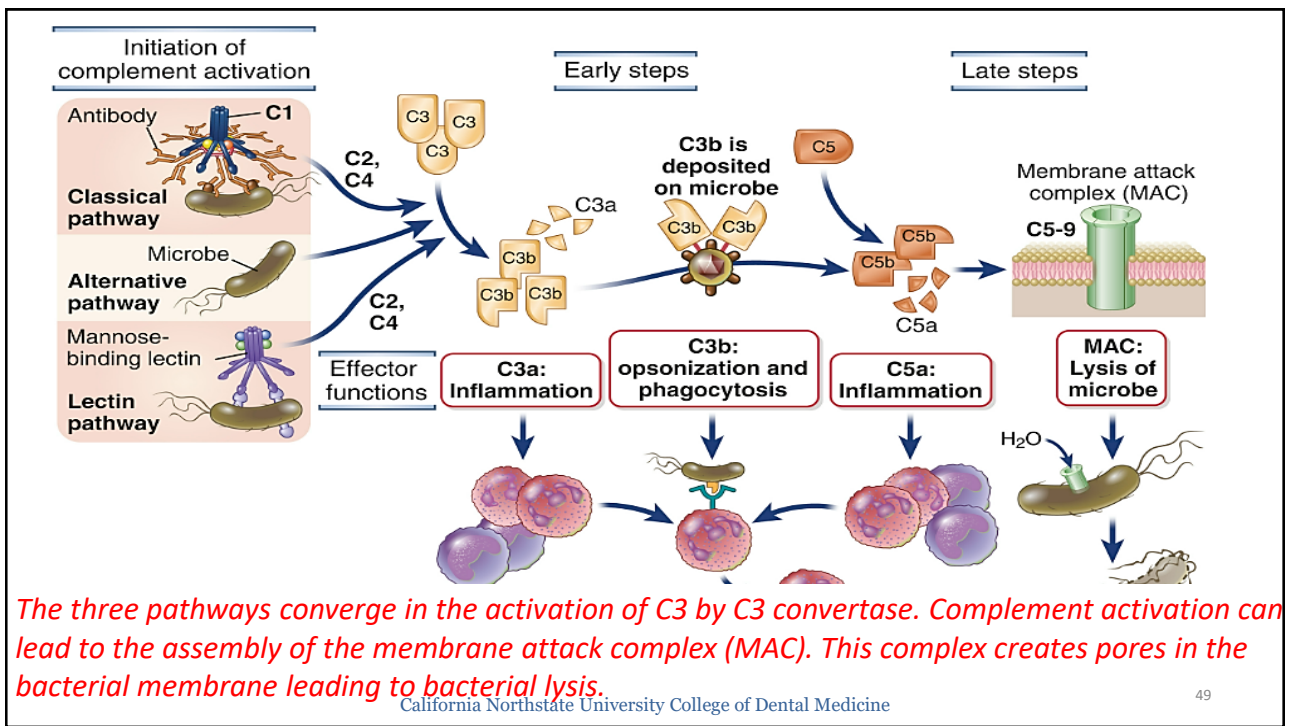
- 1- Professional Phagocytes
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Complement System

The complement system consists of 30 serum proteins that interact with one another, in a highly regulated and sequential manner, to generate products that eliminate microbes.

Proteins of the complement system are synthesized by the liver. Several complement proteins are serine proteases that can activate other members.

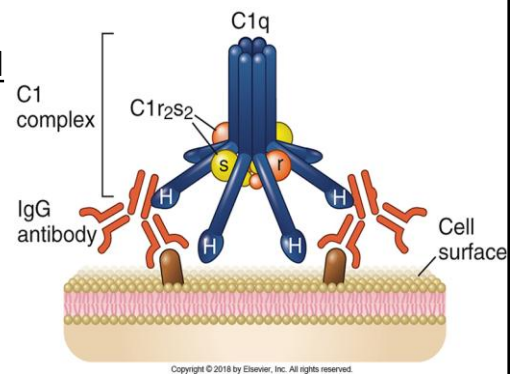
The complement system functions to induce inflammation, activate phagocytosis (via opsonization), lysis for the susceptible microbes and strengthen the humoral immune system.



Classical Pathway

Activated by Antigen-antibody interaction

- Composed of 9 complement proteins (C1-C9).
- It starts with the complement protein C1q binding to antibodies (IgM, IgG) that are bound to surface of microbe.
- This activates a proteolytic cascade that activates other members of the complement proteins (C2 and C4) and yielding the protein complex C4bC2a (C3 convertase).
- C3 convertase breaks C3 into C3a and C3b.
- The classical pathway will not be activated by soluble (non-membrane bound) antibodies.



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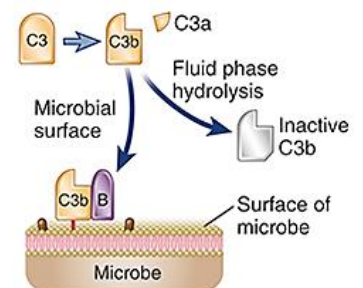
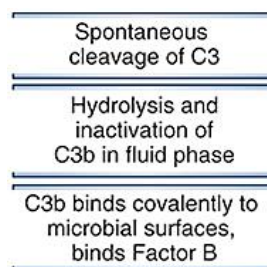
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Alternative Pathway

Activated by microbe recognition

Activated by the direct recognition of microbial PAMPs with the complement protein C3.

C3 convertase will cleave C3 protein to C3a and C3b.



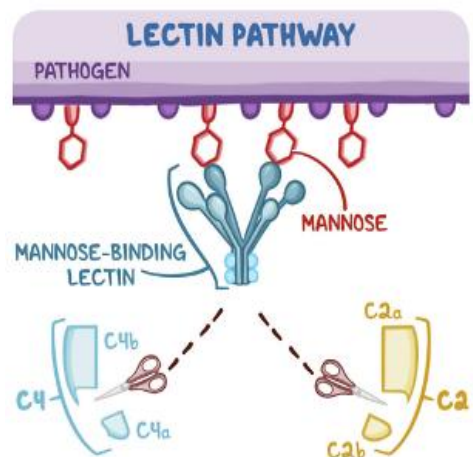
- **C3b** fragment serves as an opsonin to promote phagocytosis of the microbes.
- **C3a** is released and stimulates inflammation (chemoattractant for neutrophils) and induces mast cell degranulation to release histamine

Lectin Binding Pathway

Activated by microbial mannose

Triggered by recognition of microbial mannose residues by mannose-binding lectin (one of the plasma proteins made by the liver).

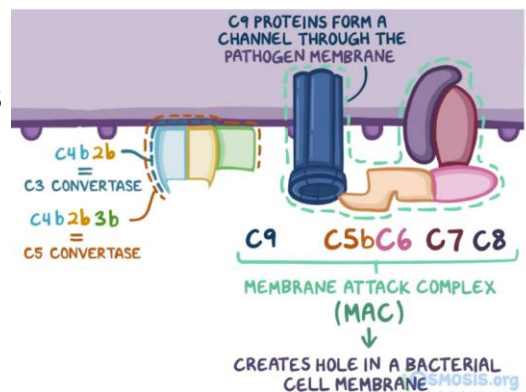
Following this binding, the mannose-associated serine protease (MASP) initiate downstream proteolytic steps to cleave C2 and C4, forming C3 convertase and cleaving C3 to C3a and C3b.



Convergence to C3 and C5

The activated complement pathway converges to

- 1- C3 convertase: cleaves C3 producing C3a and C3b.
- 2- C5 Convertase: cleaves C5 to C5a and C5b.
- 3- C5b binds C6,7,8 & 9 and initiates MAC (membrane attack complex).
- 4- MAC stimulates cytolysis by making small pores where it is assembled. This allows water and nutrient escape leading to cell death.



Outcome of Complement System

- 1- Stimulation of cytolysis by the MAC.
- 2- C3b: Opsonization of the microbe, C3b coats the microbial surface which increases of recognition of pathogens by phagocytes and hence phagocytosis.
- 3- C3a and C5a acts as anaphylatoxins by stimulating the degranulation of mast cells/basophiles and the release of proinflammatory cytokines like histamine to stimulate the inflammatory response.
- 4- C3a and C5a act as chemotaxins by increasing the increase the recruitment of leukocytes.

Individuals that lack complement proteins are more prone to repeated infections and higher risk for meningitis.

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Hereditary angioedema C1 inhibitor protein deficiency

- Prevalence: 1:50,000
- Etiology: Autosomal dominant disease caused by the lack of or a dysfunctional C1-inhibitor protein. C1 inhibitor is a broad-spectrum serine protease inhibitor that inhibits several complement proteases.
- Presentation: Episodes of swelling and angioedema that affect the lips, tongue, face, hand, feet without hives or urticaria. Can also affect the airway leading to difficulty breathing. Episode can last 2-5 days. Triggers include infections, stress or trauma.
- Treatment: multiple including recombinant C1 inhibitor.



[\(PDF\) Genetic Analysis as a Practical Tool for Diagnosis of Hereditary Angioedema With Normal C1 Inhibitor: A Case Report \(researchgate.net\)](#)

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Innate Immunity in the Oral Cavity

Oral Microbiome:

- Oral streptococci are main colonizer of the biofilm (around 60 - 80%).
- Present in a symbiotic relation.
- Periodontal health is compromised upon dysbiosis of oral flora.
- Oral microbiota transplant has been proposed as a novel approach to modulate oral microbiome dysbiosis in periodontitis and suppress pathological inflammation (still in experimental phase).

- **Epithelial Barrier**

- Gingival epithelium: prevent microbial penetration into deeper tissue.
- Express TLRs to detect microbial PAMPs and produce antimicrobial peptide and cytokines to activate innate immune response.
- Junctional epithelia within the gingival sulcus and gingival crevicular fluid allow passage of immune cells (neutrophils, macrophages & others)

Saliva

- Flow rate
- Epithelial shedding. Squamous epithelia is around 55% of salivary cell content.
- PMNs (w subpopulations), around 40 % of salivary cell count and most abundant leukocyte population (95%).
- Monocytes (macrophages and dendrites) around 1.3% of the total leukocyte population.
- Other antimicrobial peptides and secretory IgA (adaptive immune response)

Outcome of Innate Immunity

- Inflammation
- Resolution
- Progress to adaptive immune response.

Inflammation is a Hallmark of Innate Immunity

In response to innate immune response and cytokine release, the inflammatory process recruits more circulating leukocytes to eliminate the offending agents and dead cells.

The main goal of inflammation is to restore homeostasis to disrupted host tissues. The inflammatory process also aid in the repair of the damaged tissue.

Kinetics of the Inflammatory responses

Initiation

1. Resident tissue cells like macrophages, mast cells, dendritic cells are activated by PAMPs or DAMPs and release inflammatory cytokines which increase leukocytes adherence by upregulating selectins and integrins.
2. Histamine and bradykinin increase vascular permeability.
3. IL-6 and TNF- α stimulate the production of C-reactive protein from the hepatocytes. In turn, the C reactive proteins activate the production of complement proteins.

Leukocyte extravasation:

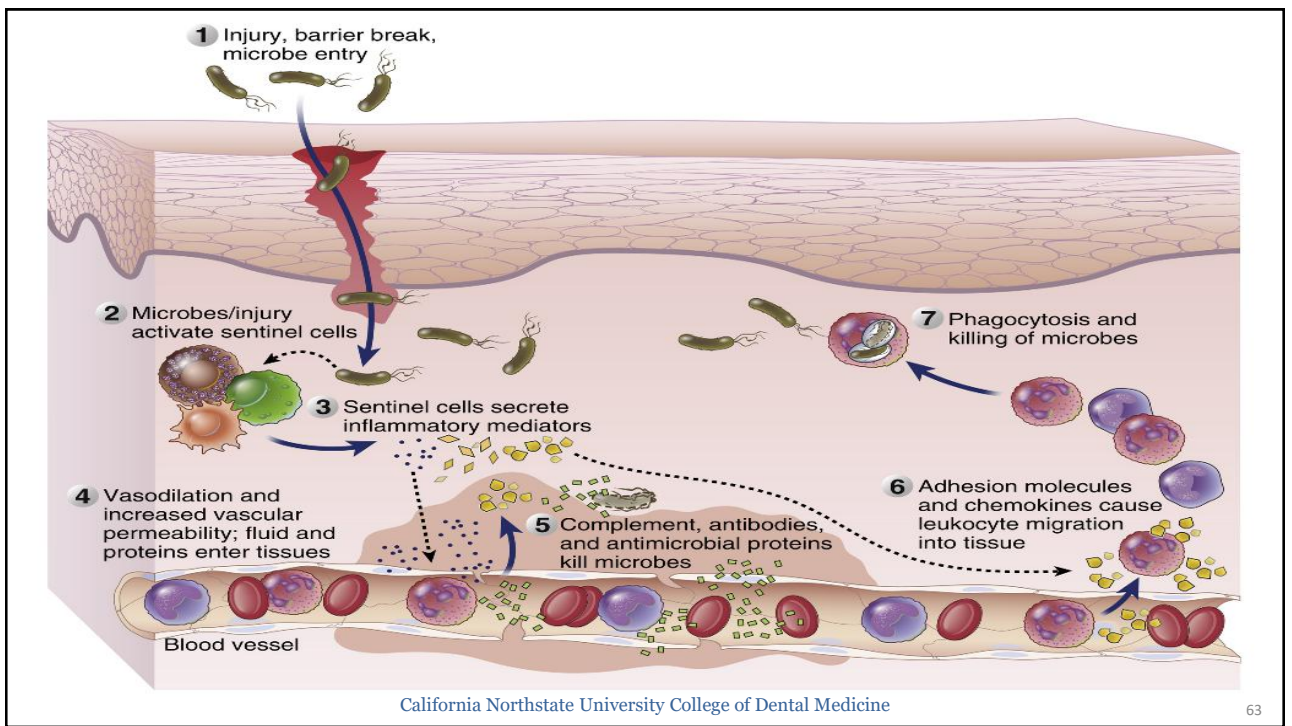
The transmigration of leukocytes through the walls of capillaries into the inflamed tissues

Elimination:

1. Recruited leukocytes are activated to phagocytose bacteria. Also, amplify the inflammatory response by producing inflammatory mediators.
2. Neutrophils are the first to be recruited to a site of infection. Macrophages participate in pathogen clearance and help in wound healing.

Clearance

Clearance of invading pathogens, damaged cells, tissue remodeling and the induction of adaptive immune responses if needed.



Big picture

- PAMPs vs DAMPs vs PRR
- Innate immune response
 - Non inducible
 - Inducible (Phagocytes, NK cells and complement activation)
- Hallmark of innate immune response
 - Acute vs chronic inflammation.
- Outcome: resolution or progression

References

